

SatQuery AI — Geospatial Analysis Report

Remote Sensing Intelligence & Multimodal Evidence Verification

Job ID:	60ec89d3-52b2-42b6-9c86-a229213e1ec5
Project ID:	e64343dc-4fb3-41c3-ac45-61635ae3beaf
Task Type:	CROMA
Status:	RUNNING
Generated:	2026-09-26 11:38:22 UTC

1. Request Specification

Query / Prompt: "Extract joint optical-SAR features."

2. Factual Findings & Observations

Finding 1 (croma_finding): Model CROMA-Base inference completed successfully.

Measurements: protocol_version: 1.0, success: True, job_id: a37a847e-1041-4c38-94b9-b4f82dc3f0b2, step_key: croma_features, attempt_id: 1, output_assets: {'feature_SAR_encodings': 'C:\\Users\\HP\\earthquery\\storage\\scratch\\a37a847e-1041-4c38-94b9-b4f82dc3f0b2\\croma_SAR_encodings.npy', 'feature_SAR_GAP': 'C:\\Users\\HP\\earthquery\\storage\\scratch\\a37a847e-1041-4c38-94b9-b4f82dc3f0b2\\croma_SAR_GAP.npy', 'feature_optical_encodings': 'C:\\Users\\HP\\earthquery\\storage\\scratch\\a37a847e-1041-4c38-94b9-b4f82dc3f0b2\\croma_optical_encodings.npy', 'feature_optical_GAP': 'C:\\Users\\HP\\earthquery\\storage\\scratch\\a37a847e-1041-4c38-94b9-b4f82dc3f0b2\\croma_optical_GAP.npy', 'feature_joint_encodings': 'C:\\Users\\HP\\earthquery\\storage\\scratch\\a37a847e-1041-4c38-94b9-b4f82dc3f0b2\\croma_joint_encodings.npy', 'feature_joint_GAP': 'C:\\Users\\HP\\earthquery\\storage\\scratch\\a37a847e-1041-4c38-94b9-b4f82dc3f0b2\\croma_joint_GAP.npy'}, output_metadata: {'model': 'CROMA-Base', 'feature_shapes': {'SAR_encodings': [1, 225, 768], 'SAR_GAP': [1, 768], 'optical_encodings': [1, 225, 768], 'optical_GAP': [1, 768], 'joint_encodings': [1, 225, 768], 'joint_GAP': [1, 768]}, 'modalities': ['optical_sentinel2_12band', 'sar_sentinel1_2band'], 'note': 'CROMA extracts joint representation features; does not generate imagery or textual answers.'}, metrics: {'duration_ms': 2442.065954208374, 'peak_vram_mb': 767.2080078125, 'peak_ram_mb': None, 'device_name': 'cuda', 'cudnn_enabled': False}, error_code: None, error_message: None

3. Operational Boundaries & Limitations

- Observations are generated based on local hardware inference (RTX 5060 Laptop GPU).
- ChangeFormer outputs represent 2D building change masks trained on LEVIR-CD.
- UPerNet semantic segmentation reflects generic ADE20K demo proxy classes unless domain-trained.
- Satellite pixel coordinates are not mapped to geographic bounds unless explicit CRS metadata is verified.
- Analyst confirmation is recommended prior to operational deployment.